

Where Should Promotional Investment Go?

Price elasticity and promotional ROI across 524,950 store-week transactions in U.S. grocery retail

77

STORES

58

SKUs · 4 CATEGORIES

156

WEEKS, 2009–2011

BAG SNACKS

COLD CEREAL

FROZEN PIZZA

ORAL HYGIENE

Every promotion looks successful in volume. Not every promotion pays for itself — this analysis separates the two.

Two Models, One Question

Does price move volume, and does that volume actually pay for the discount?

1

PRICE ELASTICITY

$\log(\text{UNITS}) \sim \log(\text{PRICE}) + \text{FEATURE} + \text{DISPLAY}$
 $+ \text{factor}(\text{STORE}) + \text{factor}(\text{MONTH})$

- Fit per product (55 SKUs) — not pooled. Different SKUs have different price levels and true elasticities; pooling them lets the price coefficient absorb between-product differences.
- The coefficient on $\log(\text{PRICE})$ IS the elasticity — log-log means no conversion is needed to read it.
- FEATURE and DISPLAY are the key modelling choice: promoted items are cheaper AND more visible. Omit the visibility controls and the price coefficient overstates true elasticity.

2

PROMOTIONAL ROI (PEA)

$\text{Portfolio ROI} = \Sigma \text{ incremental profit} \div \Sigma \text{ investment}$

- Baseline = median UNITS in non-promoted weeks, per product per store (median, not mean, so one spike can't inflate it; 10+ clean weeks required).
- Gross margin assumed at 30% on base price — stated openly, and tested at 25% and 35% for sensitivity.
- Aggregated as total profit $\div$ total investment, never the average of individual ROIs — averaging ratios overweights small events.

Both models were stress-tested before use: sheet parsing verified against raw cell counts, store-tier duplicates disclosed, margin assumption checked at 25–35%.

Price Sensitivity Varies Nearly 3x Across Categories

Median own-price elasticity, significant products only ($p < 0.05$), controlling for store and month

Oral Hygiene -1.07



Cold Cereal -1.62



Bag Snacks -1.70



Frozen Pizza -2.78



ALL FOUR CATEGORIES ARE ELASTIC ($\beta < -1$)

A price cut lifts volume more than proportionally in every category — the promotional lever works everywhere. Frozen Pizza is the most responsive; Oral Hygiene the least.

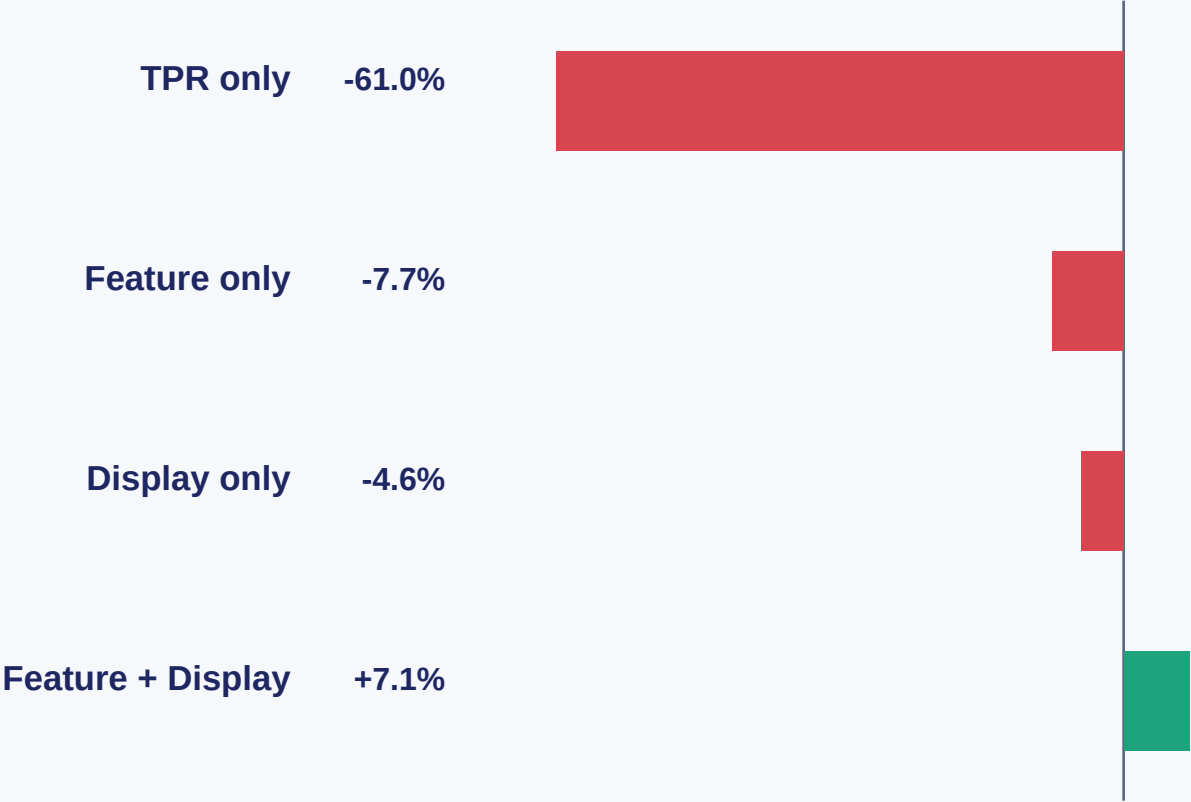
A SECOND, ROBUST FINDING

UPSCALE-tier stores are structurally less price-sensitive than MAINSTREAM in 3 of 4 categories — in Cold Cereal, 14 of 14 significant products agree on direction. Evidence for channel-differentiated pricing.

The mirror-image “VALUE tier is more price-sensitive” story did NOT hold consistently across categories — not reported as a general finding.

Half of All Promotions Are Losing Money

Portfolio ROI by mechanic = total incremental profit ÷ total investment, 101,382 events



56%

of all promotional events are TPR-only — and return –61% portfolio ROI

Only Feature + Display is value-accretive at the portfolio level. TPR-only is not just weak, it dominates volume AND destroys the most margin — only 1 in 9 individual TPR events even breaks even.

Bag Snacks and Oral Hygiene: no current mechanic has a positive portfolio ROI in either category.

Feature+Display (best) and TPR-only (worst) hold their rank at 25%, 30%, and 35% margin assumptions. Display-only / Feature-only are margin-sensitive.

THE RECOMMENDATION

Shift Promotional Investment from TPR-Only to Feature + Display

TPR-only discounting drives 56% of promotional volume and returns –61% portfolio ROI — the single largest source of value destruction in the promotional program. Feature+Display is the only mechanic with a portfolio ROI that stays positive across every margin assumption tested. Reallocating toward it — starting in Cold Cereal and Frozen Pizza, where it is already proven profitable — recovers margin without new spend.

- Elasticity confirms all four categories respond to price — the lever works. The problem is which mechanic, not whether to promote.
- Bag Snacks and Oral Hygiene have no currently profitable mechanic. Pause discretionary spend there pending a redesigned offer — don't default to TPR.
- UPSCALE-tier stores are structurally less price-sensitive. A channel-differentiated promotion depth is a lower-risk lever than a blanket price cut.

LIMITATIONS:

30% gross margin is an assumption (tested 25–35%). Promotions aren't randomly assigned — estimates carry selection bias. 2009–2011 US grocery data is directional, not a plug-in number for a different market.